

FACTORS & MULTIPLES
WHAT ARE FACTORS & MULTIPLES?





Let's Warm Up!

How can we share 12 laddoos equally with friends? Let's find out!

Skip counting helps us:

2, 4, 6, 8, 10, 12

Can you think of different ways to make equal groups?



What is a Factor?

A factor is a number that divides another number exactly, with NO leftover!

- Think of factors as "equal groups" of something
- Factors are smaller or equal to the number
 - 1 is a factor of every number
- Every number is a factor of itself



Factors Examples with Laddoos and Players



1×1



1×2



1×3



1×4



3×4



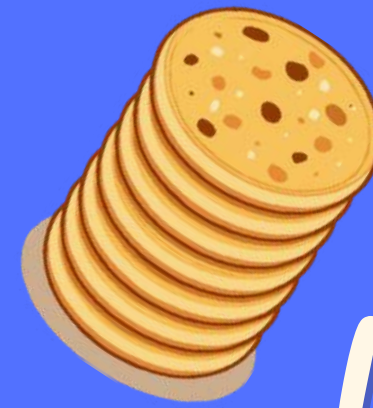
4×3

- Factors of 12 = 1, 2, 3, 4, 6, 12
- 10 cricket players → 2 teams of 5
- 16 Diwali diyas: 1×16 , 2×8 , 4×4
- 9 coconuts: 3 baskets $\times$ 3 each

What is a Multiple?

A multiple is what you get when you multiply a number by 1, 2, 3...
Think of multiples like stepping stones — each step adds the same number!
Multiples are equal to or bigger than the number.
Multiples go on forever and ever!





Multiples Examples with Rotis, Rickshaws & Coins



- Multiples of 3 rotis = 3, 6, 9, 12, 15...
- Multiples of 3 wheels on rickshaws = 3, 6, 9, 12, 15
- Multiples of ₹5 coins = 5, 10, 15, 20, 25
- Multiples of 6 biscuit packets = 6, 12, 18, 24, 30

Factors vs Multiples – What's the Difference?

FACTORS:

- Smaller or equal to the number
- Limited in number (few!)
- Found by dividing
- Example: Factors of 6 = 1, 2, 3, 6

MULTIPLES:

- Equal or larger than the number
- Unlimited (go on forever!)
- Found by multiplying
- Example: Multiples of 6 = 6, 12, 18, 24, 30...



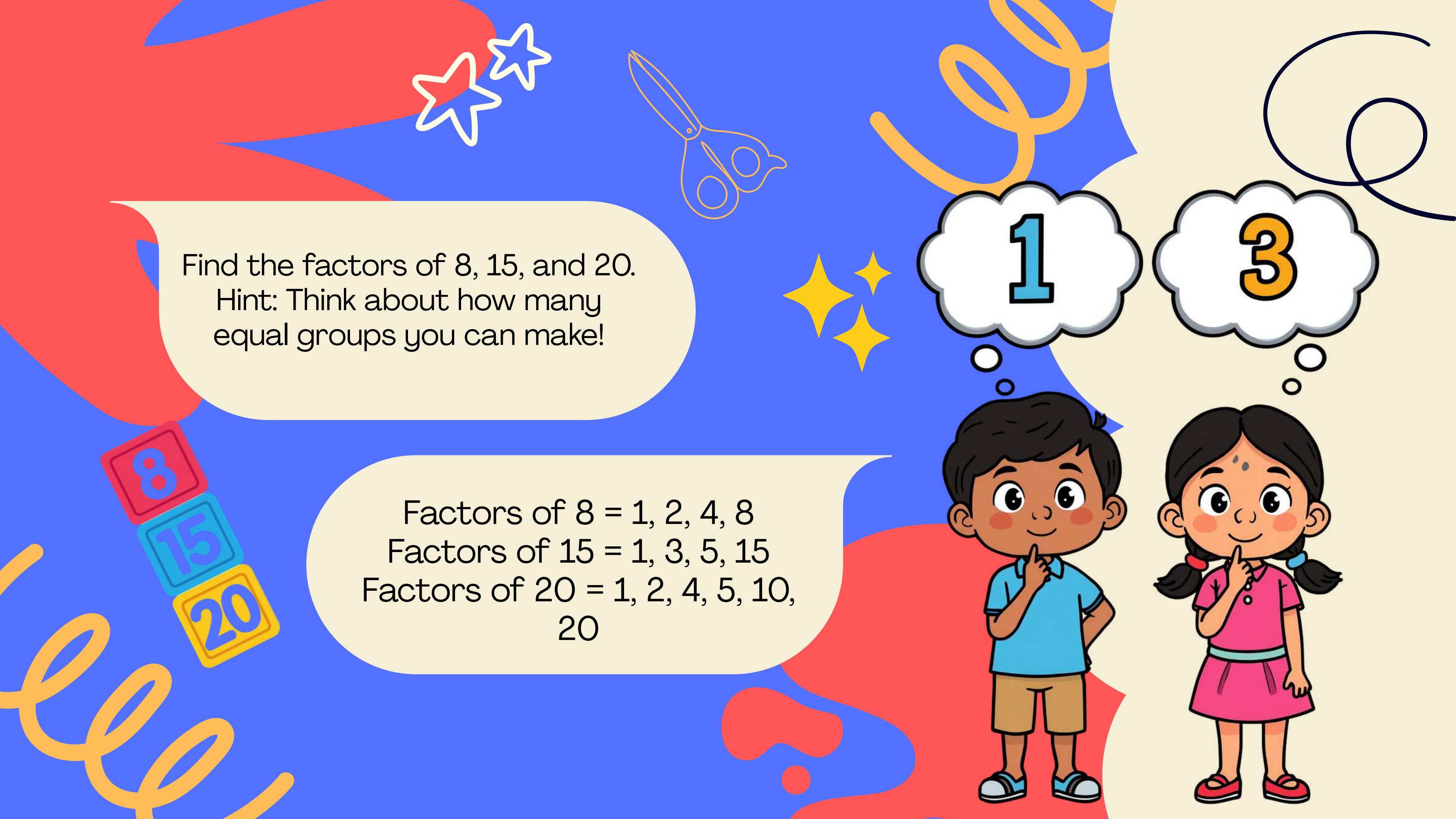
Memory Trick to Remember!

"FACTORS are FEW and SMALL —
they FIT inside!"

"MULTIPLES are MANY and BIG —
they MARCH on forever!"

Remember: Factors FIT, Multiples
MARCH!





Find the factors of 8, 15, and 20.
Hint: Think about how many
equal groups you can make!

Factors of 8 = 1, 2, 4, 8
Factors of 15 = 1, 3, 5, 15
Factors of 20 = 1, 2, 4, 5, 10,
20

1

3



Write the first 5 multiples of 4:
 $4 \times 1 = ?$, $4 \times 2 = ?$, $4 \times 3 = ?$, $4 \times 4 = ?$, $4 \times 5 = ?$

Write the first 5 multiples of 6:
Remember: Multiply by 1, 2, 3, 4, 5!



True or False? Quiz



1. Factors are always smaller than the number. ✓ True
2. Multiples stop after 10. ✗ False
3. 1 is a factor of every number. ✓ True
4. Multiples are found by dividing. ✗ False



Let's Recap Today's Learning!



- ✓ Factors divide numbers exactly with no remainder
- ✓ Multiples are numbers you get by multiplying and go on forever
- ✓ Factors are few and small; multiples are many and big

$$1+2=3$$



WHAT'S NEXT? PREVIEW OF SESSION

Next time, we'll learn how to find factors and multiples of bigger numbers!
We'll play fun games and explore more examples with your friends Arjun and Priya!
Get ready for more math adventures!



THANK
YOU!

PLANRS ACADEMY

$$1+2=3$$

